

SOFTWARE REQUIREMENTS SPECIFICATION (SRS) & SYSTEM GROUND-TRUTH DATA DICTIONARY

IEEE Std 830-1998 Compliant Specification for JGH Project Intelligence Engine

Section 1: Introduction & Executive Overview

This IEEE Std 830-1998 compliant document establishes the ground-truth technical specification for the JGH AI Intelligence Engine operating on **jghMasterDB** (MySQL 8.0, 238 tables). It defines taxonomy, schema join graphs, timestamp bounds, business metrics, and PII security controls.

Document ID:	IEEE-SRS-JGH-2026-V1.0	Target Database:	jghMasterDB (MySQL 8.0)
Compliance Std:	IEEE Std 830-1998	Total Tables:	238
Profiled Enums:	491 Columns	Security Blacklist:	16 Columns
Candidate Joins:	32,251 Edges	Revision Date:	2026-08-12
Document Status:	PENDING DOMAIN EXPERT SIGN-OFF	Licensing:	100% FOSS On-Premises Stack

IEEE 830 Terminology & Acronyms

Term / Acronym	IEEE SRS Definition & Context
AST	Abstract Syntax Tree - Structured node tree used by SQLGlot to validate SQL queries without execution.
Sargable	Search Argument Able - SQL predicate design ensuring index utilization (avoiding functions on indexed columns).
Enum	Low-cardinality discrete value set (<= 50 distinct values) mapped to domain business meanings.
PII	Personally Identifiable Information & sensitive secrets (passwords, tokens, keys) hard-blocked from LLM prompts.
RAG	Retrieval-Augmented Generation - Schema & relationship vector retrieval augmenting LLM prompts.
FOSS	Free & Open Source Software - Self-contained offline deployment stack (FastAPI, Ollama, ChromaDB, MySQL).
ERD	Entity-Relationship Diagram - Architectural visualization of database schemas and foreign key edges.
Incognito Ephemeral	Zero-history transient state mode where chat sessions leave no persistent logs.

Section 2: System Architecture & Overall Description

Product Perspective: Decoupled 6-stage execution flow:

- React + Vite Frontend UI (Chat, Data Tables, Recharts, History Drawer, Admin Dashboard)
- Security & Privacy Gateway (Input Sanitization, PII Redaction, Password/Token Scrubbing)
- Multi-Intent Router (Conversational Analytics vs Schema Tutor vs Mermaid ERD vs Executive Storyteller)
- Execution Engine (SQLGlot AST Validation -> EXPLAIN Cost Check -> Read-Only MySQL Execution)
- Response Synthesizer (Gemini-Style markdown text + tabular output rendering)
- Ephemeral Privacy Controller (Zero-history mode when is_private=true)

Section 3: System Taxonomy & User Role Ground Truth

Complete mapping of all 13 distinct user_role integer codes extracted from 29,063 profile records in the users table.

Role Code	User Count	% Share	Role Name / Title	Operational Category & Expert Sign-Off
user_role = 1	6	0.02%	Executive / Staff	Internal Operations Staff [] Approved
user_role = 2	27,703	95.32%	Retailer / Mechanic	Primary Loyalty Earners [] Approved
user_role = 3	6	0.02%	Area Executive	Field Sales Team [] Approved
user_role = 4	457	1.57%	Distributor Partner	Supply Chain Channel [] Approved
user_role = 5	573	1.97%	Wholesaler / Stockist	Supply Chain Channel [] Approved
user_role = 6	248	0.85%	Super Distributor	Enterprise Distribution [] Approved
user_role = 7	1	0.00%	System Master Admin	Platform Governance [] Approved
user_role = 8	6	0.02%	Finance Auditor	Audit & Compliance [] Approved
user_role = 10	23	0.08%	Field Executive	Field Sales & Scans [] Approved
user_role = 11	10	0.03%	Regional Manager	Regional Management [] Approved
user_role = 13	3	0.01%	Support Manager	Customer Helpdesk [] Approved
user_role = 14	1	0.00%	Sandbox / Test User	QA & Integration Testing [] Approved
user_role = NULL	25	0.09%	Unassigned Profile	Legacy Unassigned [] Approved

Section 4: Domain Semantics & Calculated Metrics

Metric Name	Target Table & Column	Proposed SQL Expression	Business Definition & Expert Sign-Off
1. Static Profile Balance	users.wallet_balance	SUM(wallet_balance)	Stored wallet balance snapshot [] Approved
2. Dynamic Ledger Balance	wallet_transaction.amount	SUM(CASE WHEN reference_type != 'withdrawal' THEN amount ELSE -amount END)	Net calculated financial balance [] Approved
3. Gross Earnings	wallet_transaction.amount	SUM(amount) WHERE reference_type IN ('cash_point', 'topup', 'incentive', 'bonus_conversion', 'referral_earning', 'credit_note')	Total monetary credits earned [] Approved
4. Total Redemptions	wallet_transaction.amount	SUM(amount) WHERE reference_type = 'withdrawal'	Total cash payouts redeemed [] Approved
5. Retailer Box Scans	sku_inventories.order_type	COUNT(*) WHERE order_type = 'Retailer' AND retailer_scanned_at IS NOT NULL	Retailer QR box scans completed [] Approved
6. Wholesaler Box Scans	sku_inventories.order_type	COUNT(*) WHERE order_type IN ('Distributor', 'Distributors') AND wholeseller_scanned_at IS NOT NULL	Wholesaler box scans completed [] Approved
7. Total Points Earned	sku_qr_points_maps.retailer_cash_points	SUM(q.retailer_cash_points + q.wholeseller_cash_points) via sku_code join	Total reward points generated [] Approved

Section 5: Relational Data Architecture & Temporal Bounds

5.1 Dual-Role Schema Join Graph

Source Entity	Join Condition & Keys	Target Entity	Business Purpose & Expert Sign-Off
users	users.id = wallet_transaction.user_id	wallet_transaction	User financial ledger history [] Approved
users	users.id = sku_inventories.status_retailer_id	sku_inventories	Retailer product scan assignment [] Approved
users	users.id = sku_inventories.status_wholesaler_id	sku_inventories	Wholesaler product scan movement [] Approved
users	users.id = sku_inventories.distributor_id	sku_inventories	Distributor inventory allocation [] Approved
sku_inventories	sku_inventories.sku_code = sku_qr_points_maps.sku_code	sku_qr_points_maps	Product UOM & Cash Points lookup [] Approved
users	users.id = withdrawal_request.user_id	withdrawal_request	Cash payout withdrawal requests [] Approved
withdrawal_request	withdrawal_request.automatic_transaction_id = automatic_transactions.id	automatic_transactions	Bank API payout execution log [] Approved
companies	companies.business_info_id = users.business_info_id	users	Enterprise business unit grouping [] Approved

5.2 Canonical Timestamp Bounds

Table Name	Timestamp Column	Data Type	Analytics Purpose & Verification
wallet_transaction	created_at	TIMESTAMP	Standard filter for monthly Earnings & Redemptions [] Verified
sku_inventories	retailer_scanned_at	TIMESTAMP	Standard filter for Retailer Box Scans in month X [] Verified
sku_inventories	wholesaler_scanned_at	TIMESTAMP	Standard filter for Wholesaler Box Scans in month X [] Verified
sku_inventories	created_at	TIMESTAMP	Inventory batch creation timestamp [] Verified
users	created_at	TIMESTAMP	User registration & onboarding timestamp [] Verified
withdrawal_request	created_at	TIMESTAMP	Cash payout request initiation timestamp [] Verified

Section 6: Profiled Enums & Security Blacklist

6.1 Target Profiled Enums Catalogue

Target Column	Distinct Database Values & Row Frequencies	Mapped Category & Expert Sign-Off
wallet_transaction.reference_type	cash_point (6.36M), withdrawal (35.9K), topup (117.5K), incentive (5.0K), bonus_conversion (3.4K), referral_earning (2.6K), credit_note (511)	Earnings vs Redemptions [] Approved
wallet_transaction.status	1 (100,000 Sampled / Active Completed)	Completed Financial Transaction [] Approved
sku_inventories.order_type	Distributor (52,106), Distributors (47,894)	Wholesaler / Distributor Scan [] Approved
sku_inventories.sources	kolkata (100,000)	Primary Warehouse Location [] Approved
sku_qr_points_maps.gride_type	FG_BIG (6,724), FG_SMALL (1,250), NULL (90)	Finished Goods Box Size [] Approved
sku_qr_points_maps.uom	B10 (3,594), B5 (3,067), B6 (64), B12 (22), B3 (1,142), B1 (153), " (22)	Package UOM Box Count [] Approved

6.2 Sensitive Column Security Blacklist

Table Name	Column Name	Data Type	Keyword Matched	Security Policy & Sign-Off
campaign_logs	device_token	varchar	token	Blacklisted from LLM [] Approved
company_controls	key	varchar	key	Blacklisted from LLM [] Approved
connection_request	auth_code	varchar	auth	Blacklisted from LLM [] Approved
employees	password	varchar	password	Blacklisted from LLM [] Approved
notification_logs	device_token	varchar	token	Blacklisted from LLM [] Approved
notifications	device_token	varchar	token	Blacklisted from LLM [] Approved
password_resets	token	varchar	token	Blacklisted from LLM [] Approved
personal_access_tokens	tokenable_type	varchar	token	Blacklisted from LLM [] Approved
personal_access_tokens	tokenable_id	bigint	token	Blacklisted from LLM [] Approved
personal_access_tokens	token	varchar	token	Blacklisted from LLM [] Approved
tripur_api_tokens	access_token	text	token	Blacklisted from LLM [] Approved
tripur_api_tokens	token_type	varchar	token	Blacklisted from LLM [] Approved
users	password	varchar	password	Blacklisted from LLM [] Approved
users	two_factor_secret	text	secret	Blacklisted from LLM [] Approved
users	remember_token	varchar	token	Blacklisted from LLM [] Approved
users	fcm_token	varchar	token	Blacklisted from LLM [] Approved

Section 7: Formal Domain Expert Sign-Off & Approval Matrix

JGH INTELLIGENCE ENGINE — IEEE STD 830-1998 FORMAL SIGN-OFF

I hereby confirm that I have reviewed the IEEE Std 830-1998 SRS specification, user role taxonomy, relational join paths, temporal date bounds, business metric definitions, and security blacklists in this document.

[] Section 3: User Roles Approved

[] Section 4: Metrics Approved

[] Section 5: Join Paths & Dates Verified

[] Section 6: Profiled Enums Approved

[] Section 7: Security Blacklist Approved

Domain Expert Name: _____

Title / Designation: _____

Signature: _____ Date: _____